# COURSE INFORMATION

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| **School/Faculty:** | Computing | **Page:** | 1 of 7 |
| **Program name:** | Bachelor of Computer Science (Software Engineering) | | |
| **Course code:** | SCSE1203 | **Academic Session/Semester:** | 20242025/2 |
| **Course name:** | Software Engineering Principles | **Pre/co requisite (course name and code, if applicable):** | SCSE1013 |
| **Credit hours:** | 3 | | |

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| **Course synopsis** | This course introduces the fundamental principles of software engineering, focusing on software development methodologies, process models, and project management techniques. Students will learn to apply software engineering concepts, interpret metrics and estimations, and develop cost-effective solutions for software projects. The course emphasizes Agile Scrum methodologies, guiding students through hands-on projects where they produce deliverables such as Project Charters, User Stories, and Low-Fidelity Prototypes. By the end of the course, students will gain essential teamwork, problem-solving, and software project management skills relevant to industry practices. | | |
| **Course coordinator** | Dr Shahliza bt Abd Halim | | |
| **Course lecturer(s)** | **Name** | **Section** | **Room no and E-mail** |
| | Dr. Shahliza bt Abd Halim | 01<br>04 | 305-08, N28<br>shahliza@utm.my |
| | Dr Jamilah bt Ahmad | 02<br>05 | 504-01, N28A<br>jamilah@utm.my |
| | Dr Norsham Idris | 03 | 305-1, N28<br>norsham@utm.my |

**Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:**

| No. | CLO | PLO (Code) | Taxonomies & generic skills | T&L methods | Assessment methods |
|---|---|---|---|---|---|
| CLO1 | Apply software engineering principle, software process models and software development methods. | PLO1 (KW) | C3 | KWL chart, Active Learning, Mind-mapping, In-class problem solving | Problem Solving: 10%<br>Quiz: 10%<br>Final: 40% |

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| Prepared by:<br>Name: Dr. Shahliza bt Abd Halim<br>Signature:<br>Date: 04 March 2025 | Certified by:<br>Name: AP. Dr. Radziah Mohamad<br>Signature:<br>Date: March 2025 |

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| CLO2 | Produce solutions to problems by interpreting suitable metrics and estimation for software development | PLO2 (AP) | C3 | Team Discussion Think-Pair-Share, Project - case study | Project Charter: 10% Low Fidelity Prototype: 12% |
| CLO3 | Present cost effective solution for software engineering project. | PLO5 (CS) | A2, CS4 | Collaborative/team discussion | 3 Phases of Sprints Daily Standup Meeting include SCRUM deliverables – Product Backlog, user story, burn down charts, burn up charts: 18% |

**Details on Innovative T&L practices:**

| No. | Type | Implementation |
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| 1. | KWL (Know-What-Learn) chart | Conducted activity to engage students in a new topic, activate prior knowledge, share unit objectives, and monitor students' learning. |
| 2. | Jigsaw-activity | Conducted collaborative group activity where students effectively teach each other (with the teacher's guidance) a given skill or procedure, topic or problem. |
| 3. | Mind-mapping | Conducted activity to visually organize information. |
| 4. | Project - case study | Conducted based on given real world problem. The students have to manage their projects by submitting specified deliverables based on the given problem. |
| 6. | Collaborative Group Discussion | Students use an online platform to interact with peers by posting thoughts or comments in response to their project discussion. |

**Weekly Schedule:**

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| **Week 1**<br>17 – 23 March 2025<br>***//Online***<br>*PH 23 March (Sun) Sultan Johor Birthday* | **Topic 1: Introduction to Software Engineering**<br>● What is software engineering?<br>● Example and characteristics of software failure<br>● Types of Software<br>● Software engineering profession and ethics |

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| **Week 2**<br>24 – 30 March 2025<br>***//Online*** | **Topic 2: Features of Software**<br>● Features of Software<br>○ Sustainability in Software<br>○ Security in Software<br>○ Intelligence in Software<br>**Assessment Briefing: PS1- Software engineering application and its features (5%)**<br>**24 March 2025 Monday Lecture will be replaced with Asynchronous lecture* |
| **Week 3**<br>31 March – 06 April 2025<br>***//Online***<br>**Week 4**<br>07 – 13 April 2025<br>*PH 31 Mar-01 April2025 (Mon&Tue) Eid-Fitri* | **Topic 3: Software Life Cycle Models - Planned and Agile driven**<br>● Planning<br>● Requirements Engineering<br>● Software Design<br>● Coding and Testing<br>● Software Maintenance<br>● Plan Driven - Sotware Process Model<br>○ Waterfall<br>○ Incremental<br>● Agile Driven - Sotware Process Model<br>○ SCRUM<br>○ Kanban<br>○ Hybrid Agile Model<br>**PS1 (5%) submission**<br>**Week 3 – All sections asynchronous lecture* |
| **Week 5**<br>14 – 20 April 2025 | **Topic 4: Fundamental of Project Management (Part 1)**<br>● Metrics for Project Size Estimation<br>● Function Point Metric<br>**WBL (Day Release): Invite Talk from Industry for project problem identification**<br>**Assessment Briefing:**<br>**PS2- Project Size Estimation (5%)** |
| **Week 6**<br>21 – 27 April 2025 | **Topic 4: Fundamental of Project Management (Part 2)**<br>● Planning and Scheduling<br>● Gantt Chart<br>● Pert Charts<br>● Activity Networks and Critical Path Method<br>**Assessment Briefing: Project Charter (10%)**<br>**PS2 (5%) submission** |

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| **Week 7**<br>28 April – 04 May 2025<br><br>*PH 01 May (Thu) Labor Day* | **Topic 4: Fundamental of Project Management (Part 3)**<br><br>• Responsibility of Software Project Manager<br>• Skill necessary for Software Project Manager<br>• Organization and team structure<br>• Identify Project Team – Product Owner; Scrum Master; Development Team<br>• Quiz 2: Project Management Planning and Scheduling<br><br>**02 May 2025 Monday Lecture will be replaced with Asynchronous lecture*<br>**Assessment: Quiz 1** |
| **Week 8**<br>05 – 11 May 2025 | **MID-SEMESTER BREAK** |
| **Week 9**<br>12 – 18 May 2025<br><br>**Week 10**<br>19 – 25 May 2025<br>*PH 12 May (Mon) Wesak Day* | **Topic 5: Agile Scrum Project Management (Part 1)**<br><br>• Functional Decomposition<br>• Product Backlog<br>• Ideate and Define User Stories<br>• Sprint planning and scheduling<br>• Sprint Backlog<br><br>**Submission Project Charter (10%)**<br>**Assessment Briefing: Daily Standup Meeting and SCRUM deliverables – Product Backlog, user story, burn down charts, burn up charts (18%)**<br><br>**Week 9 12 May 2025 Monday Lecture will be replaced with Asynchronous lecture* |
| **Week 11**<br>26 May – 01 June 2025 | **Topic 5: Agile Scrum Project Management (Part 2)**<br><br>• Burndown and Burn up Chart<br>• Daily standup meeting<br>• Sprint Review<br>• Sprint Retrospective<br><br>**Assessment: Quiz 2**<br>**WBL (Day Release): Invite Talk from Industry – Agile Project Management in the Industry** |
| **Week 12**<br>02 – 08 June 2025<br>*PH 03 June (Mon) YDP Agong's BD* | **Agile Scrum Project Implementation –Sprint #1**<br><br>**Daily Standup Meeting and SCRUM deliverables (6%)**<br>**Assessment Briefing: Low Fidelity Mock-up Prototype (12%)**<br>**03 June 2025 (Mon) – Section 01 & 02 Asynchronous* |

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| **Week 13**<br>09 - 15<br>June 2025 | **Agile Scrum Project Implementation –Sprint #2**<br><br>**Daily Standup Meeting and SCRUM deliverables (6%)** |
| **Week 14**<br>16 - 22<br>June 2025 | **Agile Scrum Project Implementation –Sprint #3**<br><br>**Daily Standup Meeting and SCRUM deliverables (6%)** |
| **Week 15**<br>23 -<br>29 June 2025 | **Low fidelity prototype demo (12%)** |
| **Week 16 – 18**<br>30 June -<br>20 July 2025 | **REVISION WEEK AND FINAL EXAM** |

**Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):**

Communication with stakeholder during requirements elicitation
Analytical and designing during software modelling
Team working using SCRUM

**Student learning time (SLT) details:**

| Distribution of student Learning Time (SLT) Course content outline | Guided Learning (Face to Face) | | | | Teaching and Learning Activities: Guided Learning Non-Face to Face | Teaching and Learning Activities: Independent Learning Non-Face to face | TOTAL SLT |
|---|---|---|---|---|---|---|---|
| **CLO** | **L** | **T** | **P** | **O** | | | |
| CLO1 | 21h | | | | 20h | 19h | 60h (minus Test 2h) |
| CLO2 | 17h | | | | 17h | 12h | 46h |
| CLO3 | 4h | | | | 4h | 3h | 11h |
| **Total SLT** | **42h** | | | | **41h** | **34h** | **115h** |

L: Lecture, T: Tutorial, P: Practical, O: Others

| Continuous Assessment | | PLO (Code) | Percentage | Total SLT |
|---|---|---|---|---|
| 1 | Problem Solving 1 | PLO1 (KW) | 5 | As in CLO1 |
| 2 | Problem Solving 2 | PLO1 (KW) | 5 | As in CLO1 |
| 3 | Quiz 1 | PLO1 (KW) | 5 | As in CLO1 |
| 4 | Quiz 2 | PLO1 (KW) | 5 | As in CLO1 |
| 5 | Project Charter | PLO2 (AP) | 10 | As in CLO2 |
| 6 | Daily Standup Meeting and SCRUM deliverables | PLO5 (CS) | 18 (3 sprints) | As in CLO2, CLO3 |
| 8 | Low Fidelity prototype | PLO2 (AP) | 12 | As in CLO2 |
| 9 | | | | |
| Summative Assessment | | | Percentage | Total SLT |
| 10 | Final Exam | PLO1 (KW) | 40 | 3h |
| **Grand Total** | | | **100** | **120h** |

**Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):**

N/A

**Learning resources:**

**Main references**

Cobb, C. G. (2023). The project manager's guide to mastering agile: Principles and practices for an adaptive approach (2nd ed.) Management Concepts.

**Additional references**

**Learning Management System (LMS)**

**http://elearning.utm.my**

**Academic honesty and plagiarism:**

Project deliverables - Project Charter; Daily Standup Meeting and SCRUM deliverables; Low Fidelity prototype are sequentially delivered and formatively graded. Copying of work (texts, simulation results etc.) from other teams or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

**Other additional information:**

All students must adhere to the following to pass a computing course:

i. Pass both **continuous assessment AND final assessment**

ii. Get at least **minimum pass marks following UTM regulation (40%)**

- The passing mark for **Continuous Assessment** is **30%** of continuous marks.
- The passing mark for the **Final Assessment** is **20%** of the final marks.
- If a student **does not achieve the passing mark** for continuous assessment or the passing mark for the final assessment, then the student's **overall final mark will be set below the UTM passing mark**.

**Disclaimer:**


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